

WorldAnchor: Evaluating and Improving 3D Consistency in Video World Models

Odyssey 2026
P035201

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Research Background

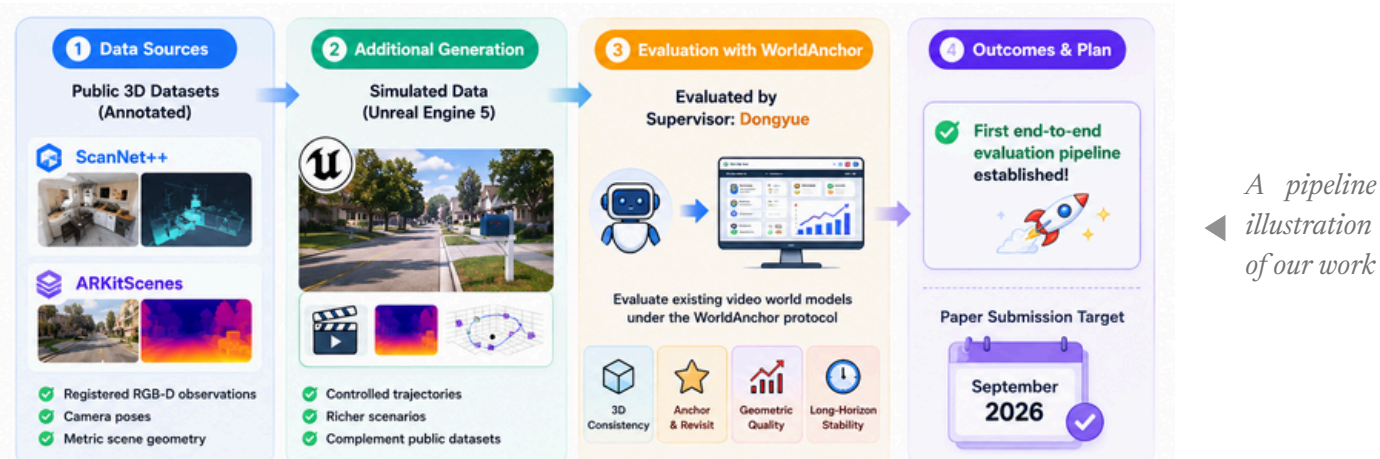
Recent **video world models** have demonstrated strong capabilities in long-horizon, camera-controllable scene generation, creating new possibilities for embodied AI, simulation, and interactive environments.



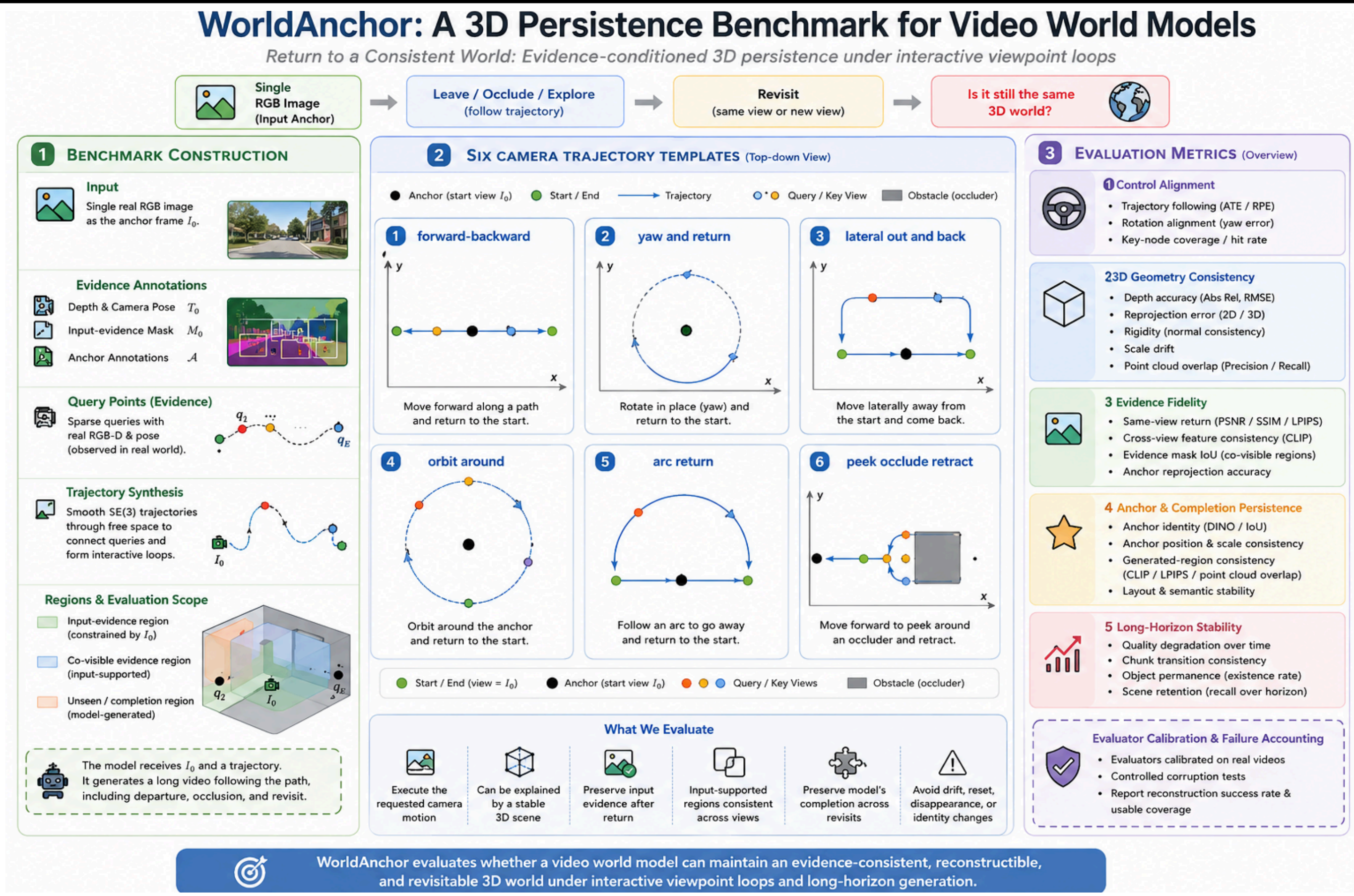
However, generating visually plausible videos does not necessarily imply that the model maintains a **coherent underlying 3D world**. During extended camera motion, viewpoint changes, occlusion, or revisits, current models may exhibit geometric drift, scale changes, layout resets, object disappearance or duplication, and inconsistent scene reconstruction. Existing benchmarks have increasingly incorporated pose estimation, depth consistency, reprojection, reconstruction, and revisit-based evaluation. Nevertheless, these approaches often evaluate different aspects of consistency **in isolation** and may **conflate camera-control failure** with failures of world persistence. More importantly, under single-frame initialization, previously unseen regions are not uniquely determined by the input image, making direct comparison with hidden ground truth potentially inappropriate. This motivates the study of evidence-conditioned 3D persistence: whether a generated world **remains consistent with the observable input evidence**, can be explained by a **stable 3D structure**, and **preserves both observed content and model-generated completions after departure and revisit**. WorldAnchor addresses this gap through interactive viewpoint loops and observability-aware evaluation of long-horizon world consistency.

Research Result

We have completed the initial annotation and benchmark case construction using public 3D datasets, including **ScanNet++** and **ARKitScenes**, which provide registered RGB-D observations, camera poses, and metric scene geometry. In parallel, we are generating additional simulated benchmark data in **Unreal Engine 5** to support controlled trajectories and richer evaluation scenarios. The annotated public cases are now being used by our project supervisor, Dongyue, to evaluate existing video world models under the WorldAnchor protocol. These efforts establish the first end-to-end evaluation pipeline for the project, with a paper submission targeted for **September 2026**.



Introduction



WorldAnchor asks a simple question: *after a model leaves the initial view, explores or becomes occluded, can it return to the same 3D world?* Starting from a single real RGB image, we construct **evidence-aware interactive trajectories** that explicitly contain departure, viewpoint change, occlusion, and revisit. Sparse RGB-D observations, camera poses, input-evidence masks, and anchor annotations provide evaluation support without requiring a complete ground-truth video. **Six trajectory templates** stress complementary forms of persistence, ranging from forward-backward motion and yaw returns to orbiting, curved loops, and occlusion-retraction. We then evaluate the generated rollout at multiple levels: whether the requested camera motion is executed, whether the video can be explained by a stable 3D scene, whether input-supported appearance and anchors are preserved across views, whether newly generated regions remain consistent when revisited, and whether these properties degrade over long horizons. Crucially, WorldAnchor **separates input-evidence preservation from generated-completion persistence**: regions constrained by the input are evaluated against real evidence, while previously unseen regions are evaluated against the model's own first completion upon revisit. Together, these tests measure whether a video world model produces not merely a coherent sequence of frames, but a reconstructible, evidence-consistent, and revisitable 3D world.

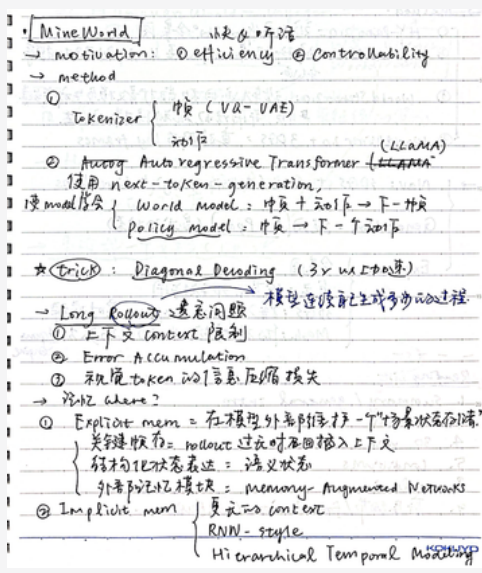
What did I do

i. Pre-reading

1. Recommended Starting Points

Priority	Resource	Type	Why read it
A	Awesome From Video Generation to World Model	Awesome list	Its taxonomy is closest to this project, from video foundation models to full-fledged, interaction, planning, datasets, and evaluation.
A	CVPR 2025 Tutorial: From Video Generation to World Model	Tutorial	A quick way for undergraduate students to build a map of the field.
A	Simulating the Visual World with Artificial Intelligence: A Roadmap	Roadmap / Survey	A systematic overview of the capabilities, data, and evaluation problems for interactive visual world simulations.
A	Game: Generative Interactive Environments	Paper	An important starting point for video-based interactive world models.
A	Game 3: A New Frontier for World Models	Official blog / tweet	Understand the target form of a real-time interactive video world model.
A	Mindset	Official blog / tweet	Contrast with Game 3 and understand the infeasible, persistent 3D world direction.
B	Awesome Video World Models	Awesome list	Additional recent projects, surveys, datasets, and benchmarks.

List of papers in the provided reading list



One page of my paper-reading notes

Reviewed at least six papers from this [reading list](#) and take structured notes to develop a broad understanding of the field, including the design principles, capabilities, and evaluation protocols of current world models and benchmarks.

ii. Reproduced current world model

Table of Contents

- First-time Setup
 - 1. SSH login
 - 2. Download dependencies on the login node
 - 3. Apply GPU node
 - 4. Initialize Conda
 - 5. Create virtual environment on the compute node
 - 6. Download pip
 - 7. Download torch (with a confirmed version)
 - 8. Download the rest of dependencies (aside from flash_attn)
 - 9. Double check torch and CUDA
 - 10. Download flash_attn
- Usual login (with setup completed)
 - 1. SSH login
 - 2. Apply GPU node
 - 3. Initialize Conda and activate the env
 - 4. Start working
- Download LingBot model weights
 - 1. Enter trux
 - 2. Download model weights inside trux
- Single GPU smoke test on LingBot
 - Run 1
 - Run 2
 - Run 3

Table of contents page of my setup documentation

Single GPU smoke test on LingBot

Purpose: Use the default picture in the original repository to do a quick smoke test to check everything can work.

```
torchrun --nproc_per_node=1 generate.py \
--task 12w-4540 \
--size 400x312 \
--ckpt_dir lingbot-400x312-base-ckpt \
--image_examples/001/image.jpg \
--action_path examples/001 \
--action_starting "w 30 s 10 s 10 s 10" \
--allow_act12w \
--sample_size 20 \
--nsteps_size 1 \
--15_cnv \
--prompt "A robot walking through a futuristic city street at night, neon lights reflecting on the wet pavement."
```

Note: I changed the parameters a bit as I can only have access to a single GPU, as opposed to what is required in `README` (8 GPUs).

Results: It takes around 9 minutes to generate a 2-second video under this setting.

Important

You can change the parameter `--sample_steps 20` to adjust the time to generate the video.

Detailed documentation on how to run a smoke test of LingBot on a single GPU

Reproduced **LingBot**, a representative video world model, on the **SoC Computing Cluster** and documented the **end-to-end setup** procedure, including cluster access, environment configuration, and dependency installation. (Refer to the document [here](#))



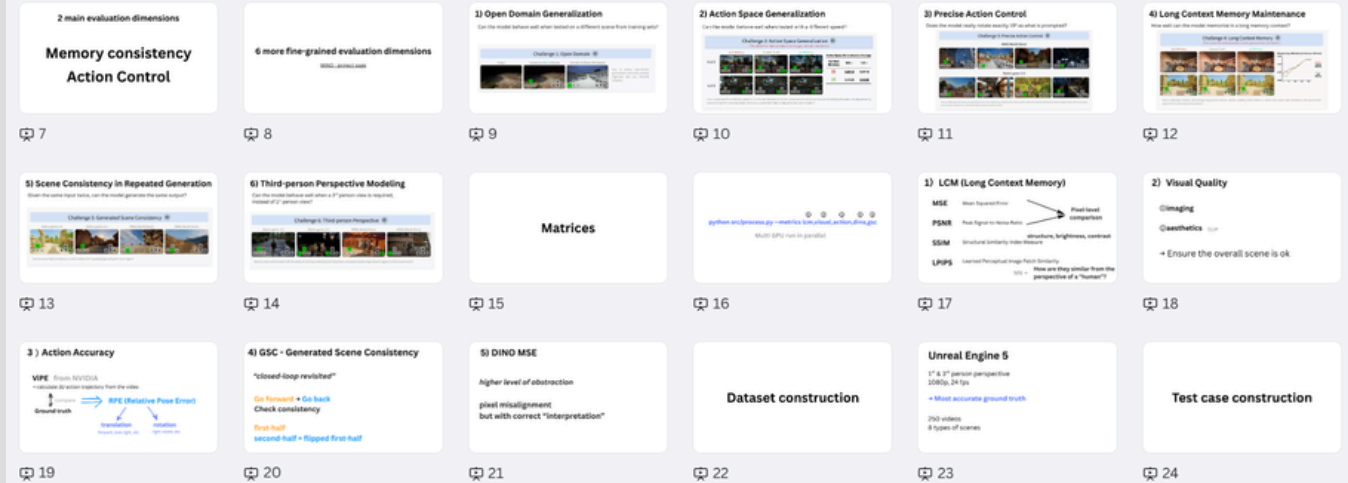
Screenshot taken from the 1st video



Screenshot taken from the 2nd video

Successfully generated **two test videos** from a single input image and corresponding scene-description prompts, while recording detailed implementation notes to support future reproducibility.

iii. Reproduced current benchmark



Analyzed and reproduced **MIND**, a representative benchmark for world models, and developed a **presentation** summarizing its methodology, evaluation framework, and key findings to facilitate discussion and knowledge sharing within the team.

LCM — Long Context Memory (vs. Ground Truth)

Metric	Score	Reference	Assessment
PSNR ↑	12.59 dB	>50 good, >20 acceptable	✗ Poor
SSIM ↑	0.411	Closer to 1 is better	✗ Low
LPIPS ↓	0.600	Closer to 0 is better	✗ High divergence
MSE ↓	0.058	Closer to 0 is better	✗ High

Input image and action series fed to LingBot

Tested and evaluated LingBot using an input image and the evaluation metrics provided by MIND. Generated a video under the benchmark setting and conducted a detailed analysis of the resulting performance and evaluation outcomes.

iv. Generated benchmark proposal

3. Overall Framework Design

The design follows the principle of **maximizing evidence**, where each frame contains as much as possible, covering the underlying physical world to avoid the model's hallucinations.

3.1 Tier 1: Pure Rigid-Body Scenes (Baseline Calibration Layer)

These scenes are designed to be simple and unambiguous, allowing for easy comparison with ground truth. They include:

- 1.1 Mailbox
- 1.2 Pumpkins
- 1.3 Metal Barrels
- 1.4 Picnic Table
- 1.5 Roadsign

3.2 Tier 2: Human Present but Static (Breathing, Blinking, etc.)

These scenes are designed to be more complex, requiring the model to handle human presence and static objects. They include:

- 2.1 Human Present but Static (Breathing, Blinking, etc.)
- 2.2 Human Present but Static (Breathing, Blinking, etc.)
- 2.3 Human Present but Static (Breathing, Blinking, etc.)

Screenshots taken from my initial proposal, and two possible datasets that I thought we might use to construct our benchmark

Developed an **initial research proposal** focused on **non-rigid body video generation**, exploring potential problem formulations and evaluation directions. Although the proposal was ultimately not selected as the team's final research direction, the process helped refine the project scope and inform subsequent proposal development.

WorldAnchor: A 3D Persistence Benchmark Proposal for Video World Models

1. Proposal Summary

WorldAnchor evaluates whether a video world model can maintain a 3D world that is consistent with the observable input evidence, can be explained by a stable 3D scene, and whether input-supported appearance and anchors are preserved across views, whether newly generated regions remain consistent when revisited, and whether these properties degrade over long horizons. Starting from a single real RGB image, the model generates a long video under a controlled trajectory, and the generated video is evaluated against the input evidence, camera poses, and other required annotations, enabling downstream geometric analysis and benchmark evaluation.

The benchmark is designed to be **simple and unambiguous**, allowing for easy comparison with ground truth. It includes:

- 1.1 Mailbox
- 1.2 Pumpkins
- 1.3 Metal Barrels
- 1.4 Picnic Table
- 1.5 Roadsign

A single real input image + a simple guided loop trajectory + input evidence masks + camera poses + 3D persistence evaluation under the generated world.

v. Generated our dataset

Generated **30 simulated videos** using **Unreal Engine 5**, spanning **five distinct anchors** and **six camera-trajectory patterns** for each anchor: **forward-backward**, **lateral out-and-back**, **arc return**, **orbit-around**, **yaw return**, and **peek-occlude-retract**. For each video, the corresponding evaluation **metadata** were also exported, including depth maps, camera poses, segmentation masks, and other required annotations, enabling downstream geometric analysis and benchmark evaluation. (Refer to the [code](#) and [the videos with metadata](#) here)

WorldAnchor Synthetic Dataset (Unreal Engine 5)

30 Simulated Videos = 5 Distinct Anchors × 6 Camera-Trajectory Patterns per Anchor

Generated with Unreal Engine 5

Five Distinct Anchors

- 1. mailbox
- 2. roadsign
- 3. metal barrel
- 4. picnic table
- 5. pumpkin

Six Camera-Trajectory Patterns

- 1. forward-backward
- 2. lateral out-and-back
- 3. arc return
- 4. orbit-around
- 5. yaw return
- 6. peek-occlude-retract

Per Anchor

- 6 trajectory patterns
- 6 videos
- Long-horizon loops with revisits
- Rich metadata for evaluation

Exported Evaluation Metadata (per video)

- RGB Frames
- Depth Map
- Camera Pose (trajectory)
- Semantic Segmentation
- Instance Mask
- 3D Bounding Boxes
- Other Annotations

Enables:

- Geometric Analysis
- 3D Consistency Evaluation
- Anchor & Revisit Assessment
- Benchmark Leaderboard

These 30 videos with rich metadata form the core of WorldAnchor, supporting comprehensive evaluation of 3D persistence under interactive viewpoint loops.

Overview of the data construction by using UE5

anchor 1: mailbox

anchor 2: pumpkins

anchor 3: metal barrels

anchor 4: picnic table

anchor 5: roadsign

Files structured for one trajectory / one anchor

- depth
- instance_mask
- logs
- rgb
- worldanchor_public
- alignment_report.json
- batch_case_status.json
- camera_frames.json
- capture_manifest.json
- clearance_report.json
- instances.json
- instances_source.json
- preview.mp4
- render_status.json
- scene_markers.json
- trajectory_config.json
- trajectory_spec.json

Note: The 3D assets and environments were **not created from scratch**. Instead, we used freely available environments from **Fab**, ensuring that all selected assets were distributed under appropriate licenses permitting their use in this project.

Scan to connect :-D

